SCADA Hub

Work Item Punchlist · ADO Submission · 2026-05-16

OVERVIEW

8 study guide sites covering Modbus, OPC UA, DNP3, IEC 61131-3, PID Controllers, SEL RTAC, Ignition SCADA, and Wireshark. Each guide is a React/Vite SPA deployed to GitHub Pages, with quizzes, flashcards, a Code Lab, and a downloadable course PDF. The hub landing page (scada-hub.vercel.app) links all 8 guides.

FEATURE WORK

1. **Quiz Question Hints** — **HIGH**
   Add a `hint` field to every quiz question across all 8 guides' `quizzes.js` data files — each hint points to WHERE the answer lives (chapter name, section heading, flashcard ID). Wire up a "Hint" button in `Quiz.jsx` that reveals the source on click. Touches all 8 guides × 600–900 questions each.
2. **Certification Relevance Sidebar Panel** — **MED**
   Build a new sidebar button per guide (modal or slide-out panel) with the cert details: name, exam body, link, description, and how each course chapter maps to exam domains. This content got cut from the syllabus PDF — the sidebar panel is its permanent home. Touches all 8 guides.
3. **DNP3 Study Guide LaTeX PDF** — **MED**
   Write, compile, and deploy the DNP3 full-length study guide PDF (same scope as the Modbus PDF). Drop it at `site/public/study_guide.pdf`. Right now the sidebar PDF button returns a 404. DNP3 guide only.
4. **Wrong-Answer Tracking + Review Missed Mode** — **MED**
   Track which question IDs a user missed per chapter — Supabase is already wired for this. Add a "Review Missed" quiz mode that serves only the questions they got wrong. Touches all 8 guides + Supabase schema.
5. **Spaced Repetition on Flashcards** — **LOW**
   Swap out the current random shuffle for the SM-2 algorithm (~150 lines of JS). Missed cards come back sooner; cards a user has nailed surface less often. Touches all 8 guides' `Flashcards.jsx`.
6. **Timed Exam Mode** — **LOW**
   Hub-level feature: 60 questions, 90-minute timer, pulled randomly across all guides in a topic area (OT protocols, ICS security, etc.). Supabase handles scoring and the leaderboard. Touches scada-hub + Supabase.
7. **Cross-Guide "See Also" Callouts** — **LOW**
   Add "See Also" callouts inside chapters that link out to related chapters in other guides (e.g., Wireshark Modbus chapter → Modbus guide, OPC UA Security → Wireshark OPC UA chapter). Touches all 8 guides' content pages.

CONTENT GAPS — SYLLABUS VS. COURSEWORK

8. **Modbus: Prerequisites + ModRSsim2 + TCP Flashcards** — **MED**
   (a) Add a prerequisite statement to `Home.jsx` / Intro chapter. (b) Call out ModRSsim2 in the TCP or Required Software section of the web guide. (c) Beef up the TCP chapter flashcards — currently 3; the syllabus implies 5+, covering transaction ID pipelining and Unit ID in gateways.
9. **OPC UA: Sessions/Channels Chapter + Discovery Section** — **MED**
   (a) Add a Sessions & Channels section to `Architecture.jsx` — token renewal, reconnection retry, ActivateSession flow. (b) Add a Discovery section: GetEndpoints, FindServers, GDS pattern, certificate trust flow. (c) Get open62541 into at least one chapter page — right now it only shows up in quiz questions.
10. **IEC 61131-3: PLCopen FB Library + Vendor Portability** — **MED**
    (a) Add a section on PLCopen standard FB signatures (TON/TOF/CTU), instance management, and how to navigate a library. (b) Add a vendor portability section on what breaks when you move code across Siemens/Beckhoff/SEL. (c) Add a table showing which IEC version applies to which platforms.
11. **PID: Scilab Lab Exercises + Laplace Prerequisites** — **MED**
    (a) Add Scilab step-test simulation exercises to the Process Characterization chapter. (b) Drop a note in the Digital Implementation chapter flagging the Laplace/transfer-function prerequisites — students without calculus need to know what to skip or review before they hit that section.
12. **DNP3: Prerequisites Statement in Web Guide** — **LOW**
    Add a short prerequisite note to `Home.jsx` or the Intro chapter: basic TCP/IP and Modbus familiarity assumed. All 10 topics are fully covered — everything else checks out.
13. **RTAC: Modbus Gateway Chapter** — **MED**
    Add a dedicated Modbus Gateway chapter (or at minimum a major section) covering DNP3-to-Modbus point mapping, register layout design, data type conversion rules, and how translation failures present. Syllabus Topic 4 ("Modbus Gateway") has no

matching web chapter right now. Also flesh out I/O module selection and pull the SEL-5033 virtual RTAC into the lab exercise.

14. **Ignition: OPC-UA Integration Chapter Depth** **MED**
Audit `OpcUa.jsx` against syllabus Topic 3 (OPC-UA Integration). Check that the PLC-to-live-tags workflow and Ignition-as-OPC-UA-server config are fully covered. Add a SQL prerequisite note to Home.jsx. (*Audit pending.*)
15. **Wireshark: SSH Remote Capture + tcpreplay** **LOW**
Syllabus Topic 9 (Advanced Techniques) calls out SSH remote capture and tcpreplay. Verify both are in `Advanced.jsx` and add anything that's missing. (*Audit pending.*)

BUGS & INFRASTRUCTURE

16. **Auth: Duplicate Email Signup Error** **MED**
Registering with an existing email drops a raw Supabase error on the user. Show a plain "account already exists" message with a "Reset password" link, and wire up the reset-password flow from that error state. Scope: scada-hub `AuthPage.jsx`.
17. **Quiz Reports RLS — Supabase** **MED**
Right now any logged-in user can read any other user's quiz history. Add RLS policies to the `quiz_results` table: users can only see their own rows; admin gets a bypass policy.
18. **Mobile UX Audit** **MED**
Mobile layout is rough across all 8 guides and the hub. Needs a full pass: text sizes, tap targets, sidebar behavior, LearningPath horizontal scroll on small screens.
19. **Supabase Email Confirmation Redirect** **LOW**
The confirmation email link sends users to a broken URL. Fix: update Site URL and Redirect URLs in Supabase Auth settings to point to `scada-hub.vercel.app`.
20. **LearningPath Hover Scale Bug** **LOW**
Hovering a step card fires `whileHover={scale: 1.03}` — the card pops out and text gets hard to read. Dial back or remove the scale transform. Scope: `scada-hub/site/src/components/LearningPath.jsx`.
21. **RTAC PDF Layout Overhaul** **LOW**
The RTAC study guide PDF has bad spacing, orphaned headings, and an intro style that doesn't match the Modbus PDF. Needs a formatting pass to bring it in line with the Modbus LaTeX template.
22. **OG Image Quality Improvement** **LOW**
OG images are generated with Python PIL and the system Helvetica — they work but look rough. Replace with a custom SVG or Vercel's `@vercel/og` for server-side generation at deploy time. Touches scada-hub `public/og-image.png` + all 8 guide repos.

UI / BRANDING

23. **Login Page: Company A Brand Colors** **MED**
The SCADA Hub login/auth page (`ScadaBackground.jsx` + `AuthPage.jsx`) uses orange as its primary accent. Swap it for whatever hex Company A provides. Touches `scada-hub/site/src/components/ScadaBackground.jsx` and `AuthPage.jsx`. Get the target hex before starting.
24. **Login Page: Floating Telemetry Values** **MED**
On the animated SCADA login background, the live telemetry tickers (PV kW, SOC%, RTT ms, MVA values) overlap node icons on some screen sizes. Fix: (a) reposition ticker boxes so they don't sit on top of the icon, and (b) add a slow vertical float animation to give the background a bit more life. Touches `ScadaBackground.jsx` node layout and ticker positioning logic.
25. **All Courses Sidebar Panel — Design Polish** **LOW**
The "All Courses" collapsible nav item landed in all 8 guide sidebars with guide icons and Vercel URLs. Follow-up polish: (a) highlight the current guide in the dropdown so users can tell where they are, (b) consider showing a chapter count or question count as a subtitle under each guide name.

COURSE AUDIT — MODBUS GAPS

26. **Modbus Syllabus: Add RS-485 as Standalone Topic** **LOW**
The syllabus lists 11 topics but the web guide and LaTeX PDF both treat RS-485 Physical Layer as a full standalone chapter (Ch 4). Add RS-485 as Topic 3 in the syllabus (between Data Model and RTU Mode) — brings the total to 12 topics and matches the actual chapter count.
27. **Modbus Web: Add FC08 and FC43 to Function Codes Chapter** **MED**
`FunctionCodes.jsx` only covers FC01, FC02, FC03, FC04, FC05, FC06, FC15, FC16. The LaTeX PDF (Ch 8) and flashcards both include FC08 (Diagnostics) and FC43 (Encapsulated Interface Transport / MEI). Add those two to the web chapter so all three sources stay in sync.
28. **Modbus Web: Add Exception Codes 10 and 11 to Exceptions Chapter** **MED**
`Exceptions.jsx` stops at codes 01–08. The LaTeX PDF (Ch 9) and flashcards both cover gateway codes 10 (Gateway Path

Unavailable) and 11 (Gateway Target Device Failed to Respond). Add them to the web chapter and fix the syllabus wording — it currently says "eight exception identifiers" but there are eleven.

29. **Modbus Web: Add Connection Management to TCP Chapter** LOW
`TCP.jsx` doesn't cover stale connections or multiple simultaneous connections. LaTeX Ch 7 covers both. Add a Connection Management section to the web chapter.
30. **Modbus Web: Add Poll Rate Overrun to Troubleshooting Chapter** LOW
The syllabus explicitly calls out "poll rate overruns" as a troubleshooting topic but it's nowhere in `Troubleshoot.jsx`. Add it as a symptom scenario: device busy exceptions, missed polls, sluggish HMI response — cause is too many registers per cycle or too short a poll interval for the baud rate.
31. **Modbus Web Lab: Add Gateway Exception 11 and Security Audit Scenarios** MED
LaTeX Ch 12 has Scenario 5 (Modbus TCP Gateway Exception 11) and Scenario 6 (Security Audit Finding) — neither is in `Lab.jsx`. Add both as interactive exercises or frame-decode problems to close the gap.
32. **Modbus Flashcards: Expand Security Chapter** LOW
The Ch 11 Security flashcard set has only 4 cards even though the chapter covers 5 attack vectors, 5 defense-in-depth layers, and 4 NERC CIP standards. Shoot for 8–10 cards to match the depth of other chapters.

COURSE AUDIT — OPC UA GAPS

33. **OPC UA Web: Add Dedicated Discovery Chapter** MED
The syllabus has Discovery as Topic 8 — Local Discovery Server, Global Discovery Server, FindServers, GetEndpoints, endpoint enumeration, open-discovery security risk. No `Discovery.jsx` page exists; the topic shows up briefly in `Intro.jsx` and in one flashcard (t10). Build a full web chapter covering LDS/GDS architecture, FindServers vs GetEndpoints, certificate trust during discovery, and the security risk of unauthenticated endpoint enumeration.
34. **OPC UA Flashcards: Add Troubleshooting Cards** MED
`Troubleshoot.jsx` is a solid chapter — 7 status codes, a 5-step diagnostic ladder, UA Expert workflow, Wireshark analysis — but zero flashcards cover any of it. Add 6–8 cards hitting the most common status codes (`BadCertificateUntrusted`, `BadNodeIdUnknown`, `BadUserAccessDenied`, etc.) and the diagnostic ladder sequence.
35. **OPC UA Flashcards: Add Ignition Integration Cards** LOW
`IgnitionIntegration.jsx` covers Ignition port 62541, the 5-step certificate trust workflow, and SEL RTAC as an OPC UA server — none of it appears in flashcards. Add 4–5 cards on Ignition-specific OPC UA behavior.
36. **OPC UA Web Security Chapter: Add CRL and Role-Based Security** LOW
Flashcards sec05 (Certificate Revocation Lists) and sec07 (role-based access: Observer, Operator, Engineer, Supervisor) exist but neither topic shows up in `Security.jsx`. Add subsections for CRL checking and built-in OPC UA roles to bring the web chapter in line with the flashcards.
37. **OPC UA Web Services Chapter: Add RegisterNodes** LOW
`RegisterNodes/UnregisterNodes` is a performance optimization service that appears in flashcard s08 and Level 3 quiz questions but is absent from `Services.jsx`. Add a short section on when and why to use `RegisterNodes` for high-frequency polling.

COURSE AUDIT — DNP3 GAPS

38. **DNP3 Flashcards: Add Application Layer Chapter Cards** MED
The `appLayer` chapter has a full quiz question set but zero flashcards. Add 5–7 cards covering the DNP3 Application Layer: request/response fragments, function codes (READ, WRITE, SELECT, OPERATE, DIRECT OPERATE), IIN bits, sequence numbering, and fragmentation reassembly. That's the only gap in the DNP3 audit — all 10 syllabus topics are otherwise covered.

COURSE AUDIT — IEC 61131-3 GAPS

39. **IEC 61131-3 Flashcards: Add Lab Chapter Cards** MED
The Lab & Practice chapter (`Ch 10`) has 60 quiz questions and zero flashcards. Add 6–8 cards covering the practical lab scenarios: PLC program structure, scan cycle, I/O addressing syntax, and the beginner mistakes that show up most often in the quiz scenario questions.
40. **IEC 61131-3 Flashcards: Split Languages Category by Language** MED
The `languages` flashcard category has ~47 cards crammed across four distinct chapters (Intro, Structured Text, Ladder Diagram, Function Block Diagram) — each of those chapters has 60 quiz questions but shares one thin flashcard set. Split into at least two categories: `ld-fbd` (graphical languages) and `st-il` (text languages), and expand to ~20 cards per language to match quiz depth.
41. **IEC 61131-3 Flashcards: Cover Scenario-Type Questions** LOW
The quiz data has ~80 scenario questions (`iec_*_sc` entries, ~5 per chapter) covering applied troubleshooting and design decisions. No flashcard category touches scenario-style content. Add a `scenarios` category with 10–12 cards framing the most instructive setups and how to work through them.

COURSE AUDIT — PID CONTROLLERS GAPS

42. **PID Flashcards: Add Cards for 5 Uncovered Chapters** HIGH
5 of 10 quiz chapters have zero flashcards: Closed-Loop Intro (65 Qs), Loop Fundamentals (65 Qs), P/I/D Action (65 Qs), Digital PID (65 Qs), and Troubleshooting (65 Qs). That's roughly half the quiz content with nothing to reinforce it. Priority order: (1) P/I/D Action, (2) Troubleshooting, (3) Digital PID, (4) Intro, (5) Loop Fundamentals. Shoot for 6–8 cards per chapter.
43. **PID Flashcards: Align Category Names to Chapter Boundaries** LOW
The current flashcard categories (fundamentals, tuning, dynamics, advanced, practical) don't map to the 10 syllabus chapters. A student coming off a quiz chapter can't find the matching flashcard deck. Rename or split categories so each quiz chapter has a clear flashcard counterpart.
44. **PID Flashcards: Add Lab / Simulation Chapter Cards** LOW
Ch 10 (Simulation Lab) has 65 quiz questions and zero flashcards. Add 5–6 cards covering the Scilab step-test procedure, open-loop identification, and the tuning workflow from the lab.

COURSE AUDIT — SEL RTAC GAPS

45. **RTAC Flashcards: Add Cards for 4 Uncovered Chapters** HIGH
4 quiz chapters have zero flashcards: RTAC Architecture/Intro, ACSELERATOR Architect Software, Tag Database, and Modbus Gateway — the last one being syllabus Topic 4 with no flashcard coverage at all. Shoot for 5–7 cards per chapter; prioritize Gateway and TagDB since those are the heaviest operational topics.
46. **RTAC: Add Lab Chapter Quiz Questions** MED
The Lab & Field Scenarios chapter (`Lab.jsx`) has a full web page and zero quiz questions — the only chapter across all 8 guides in that position. Write 30–40 questions covering the scenario walkthroughs on the lab page: configuration steps, fault identification, verification checks.
47. **RTAC Flashcards: Reconcile Hardware Chapter Scope** LOW
The `hardware` flashcard category currently spans three web chapters (Architecture, Software, TagDB) instead of mapping to one. Refactor: move hardware-specific cards into an `intro` category and create separate `software` and `tagdb` categories so each chapter has its own flashcard home.

COURSE AUDIT — IGNITION SCADA GAPS

48. **Ignition Flashcards: Add Cards for 4 Uncovered Chapters** HIGH
4 of 10 quiz chapters have zero flashcards: Intro/Overview (65 Qs), Ignition Gateway (65 Qs), Vision Module (65 Qs), and Alarming (65 Qs). That's 260 quiz questions with nothing backing them up. Priority order: (1) Gateway, (2) Alarming, (3) Vision, (4) Intro. Shoot for 7–8 cards per chapter.
49. **Ignition Flashcards: Reconcile Quiz/Flashcard Naming** LOW
Quiz chapters use `scripting` and `alarms`; flashcard categories use `scripts` and `alarming`. A student going from a quiz result to the matching flashcard deck hits the wrong category. Standardize the naming across both data files so IDs line up.

COURSE AUDIT — WIRESHARK GAPS

50. **Wireshark Flashcards: Add Advanced Techniques Chapter Cards** MED
Ch 9 (Advanced Techniques) has 65 quiz questions and zero flashcards. The chapter covers editcap, mergecap, tshark, dumpcap ring buffer, Lua dissector scripting, and SSH remote capture. Add 6–8 cards hitting the most testable concepts: tshark command syntax, ring-buffer capture options, the SSH remote capture one-liner, and Lua dissector registration.
51. **Wireshark Flashcards: Add Lab Chapter Cards** LOW
Ch 10 (Protocol Lab) has 65 quiz questions and zero flashcards. Add 5–6 cards covering the lab workflow: filter construction for Modbus, DNP3, and OPC UA captures; common frame-decode steps; and the ICS anomaly identification checklist from the lab exercises.
52. **Wireshark Advanced Chapter: Add tcpreplay Coverage** LOW
`Advanced.jsx` covers editcap, mergecap, tshark, and SSH remote capture but says nothing about tcpreplay (PCAP replay for ICS protocol testing), which syllabus Topic 9 explicitly lists. Add a short section: how to replay a captured PCAP against a test device, rate control, and why it matters for testing IDS/IPS rules against replayed OT traffic.

TRAINING FEATURE — LAUNCH CHECKLIST

53. **Run Training Events SQL Migration** **HIGH**
Open Supabase Dashboard → SQL Editor and run `scada-hub/supabase/20260516_003_training_events.sql`. Creates the `training_events` table, RLS policies, unique constraint on URL, and the course+date index. Has to happen before seeding or the `TrainingPanel` will break.

54. **Configure Vercel Env Vars for Training Cron** **HIGH**
In the scada-hub Vercel project settings, add these environment variables:
`CRON_SECRET` — any random string (e.g. `openssl rand -hex 32`)
`BRAVE_SEARCH_API_KEY` — free tier at `api.search.brave.com` (2000 queries/month)
`RESEND_API_KEY` — free account at `resend.com` (3000 emails/month)
`ALERT_EMAIL` — `ric3639@gmail.com` (broken URL alert destination)
`SUPABASE_SERVICE_KEY` — service role key from Supabase project settings

55. **Enable Vercel AI Gateway on scada-hub Project** **HIGH**
In Vercel Dashboard → scada-hub project → Settings → AI Gateway: flip the gateway on. This provisions the OIDC token that `api/refresh-training.js` needs to call `anthropic/claude-haiku-4.5` without a direct API key in the environment.

56. **Seed Initial Training Events** **HIGH**
After the migration runs, seed 35 verified 2026 training events and certifications across all 8 courses:
`SUPABASE_URL=...  SUPABASE_SERVICE_KEY=...  node scripts/seed-training.js`
Covers: Modbus (3), OPC UA (4), DNP3 (4), IEC 61131-3 (4), PID (5), RTAC (3), Ignition (5), Wireshark (6) + certs.

57. **Deploy All 9 Projects — Training Feature** **HIGH**
Redeploy all 9 projects to push the `TrainingPanel` component and Vercel cron config live. Run from each project root:
`cd site && GITHUB_PAGES= npm run build && cd ..  && vercel --prod --yes`
scada-hub also needs: `vercel --prod --yes` from the project root (no site build needed for the cron function).

58. **Troubleshooting Lesson Plan LaTeX PDF** **MED**
Build a multi-chapter LaTeX PDF that turns real debugging sessions into field-engineer lesson plans. Each chapter covers one class of bug or architectural decision: symptom, what was checked, root cause, fix, and what to watch for next time. Example chapter: "Missing Icon Import Crashes React App" (Ignition `Zap` ReferenceError). Generate at the end of each session and stack chapters over time. Source lives in `scada-hub/lesson-plans/`. This is separate from the per-guide study guide PDFs — it's a *developer training document*.

PROFESSIONAL DEVELOPMENT / R&D

59. **R&D: AI Agent Skills and Agentic Tooling** **LOW**
Get real reps in with agentic AI tools that actually matter for this stack — specifically Claude AI (Claude Code, agent SDK, multi-agent orchestration), OpenAI Codex, and equivalent code-generation toolchains. Figure out which capabilities can move the needle on delivery (code generation, automated testing, documentation, diagram synthesis) and build repeatable workflows around the ones that stick. Deliverable: a short internal write-up covering what was evaluated, what it can actually do in this stack, and what's worth adopting.